



Péter Juhász, PhD

Quantitative Researcher

CONTACT

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EDUCATION

2022 – 2025 PhD in Mathematics
Aarhus University, Denmark

2019 – 2021 MBA in Finance
Budapest University of Technology and Economics, Hungary

2013 – 2016 MSc in Physics
Budapest University of Technology and Economics, Hungary

2010 – 2013 BSc in Physics
Budapest University of Technology and Economics, Hungary

SKILLS

Modeling: Stochastic Models, Bayesian Statistics, Probability, Data Analysis

Programming: Machine Learning, Python, C++, SQL, Git, Monte Carlo Simulation

LANGUAGES

English

German

Hungarian

PROFILE

Quantitative Researcher with a PhD in Mathematics, specializing in stochastic modeling, machine learning, and predictive systems for financial markets. Experienced in probabilistic modeling, Monte Carlo simulation, uncertainty quantification, and statistical validation for data-driven decision-making. Currently developing intraday energy-market price prediction models and optimal liquidation strategies using machine learning, functional data analysis, and stochastic differential equations.

PROFESSIONAL EXPERIENCE

Quantitative Researcher **2026 –**
in Energy Markets & Algorithmic Trading
Aros Commodities & Aarhus University, Denmark

- Conducting quantitative research on intraday energy-market prediction and optimal liquidation strategies in collaboration with Aros Commodities.
- Developing Python-based machine learning workflows to predict intraday price movements using functional data analysis techniques.
- Modeling energy price forecast errors using jump-diffusion stochastic differential equations and Monte Carlo simulation.
- Deriving optimal liquidation strategies for intraday markets using Hamilton–Jacobi–Bellman methods.

PhD Researcher in Mathematics **2022 – 2025**
Aarhus University, Denmark

- Developed stochastic and probabilistic models to analyze complex systems under uncertainty, with focus on distributional properties and asymptotic behavior.
- Designed statistical validation methods to rigorously evaluate model assumptions and predictions against empirical observations.
- Built large-scale Monte Carlo simulations and algorithms in Python and C++ to stress-test models, evaluate edge cases, and quantify uncertainty.

CERTIFICATIONS

2021 Software Architect ISAQB

2021 Software Designer Bosch

AWARDS, GRANTS

2022 PhD Fellowship

Danish Data Science Academy
1 890 000 DKK

2021 Excellence Award

Hungarian National Bank
350 000 HUF

Machine Learning Researcher in Automated Driving

2018 — 2022

Bosch Group, Hungary

- Patented a Gaussian regression-based trajectory prediction model, improving predictive accuracy by $3\times$ and uncertainty calibration by $10\times$ over production baselines.
- Built stochastic forecasting and uncertainty quantification models for safety-critical, real-time decision-making in automated driving.
- Designed and executed large-scale data analyses and validation experiments to evaluate model assumptions and predictions against observed outcomes.
- Translated prototypes into scalable production systems with explicit reliability criteria.

Software Engineer in Performance Analytics

2016 — 2018

Ericsson Telecommunications, Hungary

- Analyzed performance and response-time distributions of large-scale distributed systems to optimize load-balancing strategies.
- Designed and implemented performance-critical algorithms for high-throughput network routing systems.

PATENTS

- P. Juhász et al. **2025**
Method and apparatus for processing data associated with at least one ultrasonic sensor
- P. Juhász. **2021**
Method and device for predicting the trajectory of a traffic participant, and sensor system

SELECTED PUBLICATIONS

- P. Juhász. **2025**
PhD Thesis: Stochastic Models of Higher-Order Networks - Point Processes and Topological Data Analysis
- C. Hirsch, B. Jahnel, and P. Juhász. **2025**
Functional limit theorems for edge counts in dynamic random connection hypergraphs
- P. Juhász. **2021**
Information propagation in stochastic networks
- P. Juhász, J. Stéger, D. Kondor, and G. Vattay. **2018**
A Bayesian approach to identify Bitcoin users